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NEPSD-202-2025

FACULTY OF SCIENCE AND TECHNOLOGY

B.Sc. (First Year) (Second Semester) EXAMINATION

APRIL/MAY, 2025

(NEP)

COMPUTER SCIENCE

SCSCMT-1151

(Data Structure-II)

(Tuesday, 15-4-2025)

Time : 10.00 a.m. to 12.00 noon

Time-2 Hours

Maximum Marks-40

N.B. :- (i)

Question No. 1 is compulsory.

(ii)

From question Nos. 2 to 5 solve any three questions.

(iii)

Figures to the right indicate full marks.

(iv)

Assume suitable data, if necessary.

(v)

Use of any electronic media such as mobile phone, digital diary, and

electronic calculator is not permitted.

1.

Attempt the following (2 marks each) :

(a)

Explain data structure operations.

(b)

Explain priority queue.

(c)

What is binary tree ?

(d)

Define data structure.

(e)

Define linked list.

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(2)

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2.

Attempt any two of the following :

(a)

Explain binary search algorithm.

(b)

What is linear array ? Discuss representation of linear array in memory.

(c)

Write an algorithm to traverse the linear array.

3.

Attempt any two of the following :

(a)

Explain Insertion sort.

(b)

Write an algorithm to delete a node from the linked list after the given

node.

(c)

Discuss the concept of garbage collection.

4.

Attempt any two of the following :

(a)

What is Stack ? Explain Array representation of stacks.

(b)

Explain arithmetic expression.

(c)

Discuss various operations of queues.

5.

Attempt any two of the following :

(a)

Explain postorder traversal of binary tree.

(b)

Write a short note on general tree.

(c)

Explain types of binary tree.

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